Week 3 Progress Report
BSE4104 Emerging Trends in Software Engineering
University Student-Support Case Agent
Group M Evening Week Ending: 19th September 2026

Work Completed Against Weekly Objectives

All five Week 3 deliverables specified in the assignment brief were completed:

Corpus/Source Register: eight public Makerere freshman onboarding documents were identified, collected, and registered with provenance metadata (title, source URL, purpose). Seven extracted successfully into plain text; the eighth (dynamic timetable portal) was excluded and noted in the register as un-embeddable static text.

RAG architecture diagram: produced and committed. Shows the two parallel flows — offline document ingestion (download, extract, chunk, embed, index) and online query handling (embed question, similarity search, context builder, grounded generation, grounding check, answer or escalation).

Working RAG pipeline with source grounding: implemented end-to-end across five scripts (download, extract, chunk, embed, retrieve) and an evaluation runner. Retrieval uses cosine similarity over Gemini embeddings with top-3 chunks above a 0.70 threshold. The grounded-answer prompt forces the model to answer only from retrieved context, cite source titles, and admit missing context rather than guessing.

15-case RAG evaluation: completed with 5 answerable, 5 partially answerable, and 5 unanswerable questions. Results: 4/5 answerable passed, 2/5 partially answerable passed, 5/5 unanswerable correctly refused. Overall 11/15 (73%).

Failure catalogue: three genuine failures documented with root cause analysis and proposed fixes.

Key Engineering Decisions

Retrieval threshold: 0.70 (conservative): Tested 0.50/0.65/0.70; chose strictest since a wrong-topic answer is worse than "I don't know." Result: zero unanswerable questions slipped through.

Chunking: paragraph-based (~1500 chars): Preferred over fixed-size token windows so chunks stay coherent and citable (no mid-sentence starts).

Embeddings: task-type paired: Used RETRIEVAL_DOCUMENT for chunks and RETRIEVAL_QUERY for questions with gemini-embedding-001, matching the model's intended use for better retrieval quality.

Pipeline: incremental saving: Saved embeddings every 10 chunks; survived a rate-limit failure (429 at chunk 71/90) by resuming next day instead of restarting.

Grounding prompt: strict constraints: Answer only from retrieved context, cite sources, admit gaps — directly meeting the grounding/unsupported-answer requirement.

Failures, Challenges and Current Response

False refusal on answerable query: "What to bring when joining" query missed the correct chunk (score < 0.70) due to wording mismatch with formal document language. Fix planned: raise top-k to 5, trial threshold at 0.68, or add metadata filtering for onboarding queries. Scheduled Week 4–5.

Corpus gap: password recovery: Only an activation guide exists, no recovery guide. Logged as a genuine content gap, not a bug. Fix: add a recovery guide if available, or a fallback escalation chunk.

Dynamic content out of scope: Registration deadlines live on a portal, can't be statically embedded. Resolved by escalating such queries to a support ticket, per the charter's escalation rule.

Rate-limit hit during embedding: 429 error at chunk 71/90. Incremental saving meant no lost progress; pipeline resumed cleanly next day.

Links and Evidence

GitHub repository (branch: rag-evaluation):

https://github.com/SumayyaN/BSE41014_capstone_group_m_ee

Individual Contribution Summary

Aryampa Joshua: Corpus assembly and Source Register

Nakaye Hildah: RAG architecture diagram

Namata Sumayya: RAG pipeline (download, extract, chunk, embed, retrieve)

Mushabe Moses: 15-case evaluation, failure catalogue, and this progress report